Chapter 14. Applications to Group Theory

Covering spaces of graphs, fundamental groups of graphs, subgroups of free groups

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Covering space theory gives clean proofs of algebraic facts about free groups, by realizing free groups as fundamental groups of graphs.

14.1 Covering spaces of a graph

A **linear graph** is a space X that is a union of arcs (edges) meeting only at endpoints (vertices), with the coherent topology (a set is closed iff its intersection with each edge is closed). Graphs are Hausdorff, locally path connected and semilocally simply connected. A **tree** is a connected graph containing no simple closed curve (no cycle).

Theorem 14.1

A covering space of a graph is a graph: if $p : E \to X$ is a covering and X a graph, the lifts of edges and vertices make E a graph, and p maps edges homeomorphically onto edges.

Proof. Each edge is simply connected, so it lifts homeomorphically through every point of its preimage (general lifting lemma); the coherent topology on E follows from that on X and local homeomorphism. ■

14.2 The fundamental group of a graph

Lemma 14.2

Every tree is simply connected; indeed every finite tree, and any tree, deformation retracts onto a vertex (shrink one terminal edge at a time for finite trees; use the coherent topology in general).

Theorem 14.3

Every connected graph X has a maximal tree T (containing all vertices). If the edges not in T are $\{a_\alpha\}$, then $\pi_1(X)$ is free with one generator for each a_α. For a finite connected graph with V vertices and E edges, $\pi_1(X)$ is free of rank $1 - \chi(X) = E - V + 1$.

Proof. A maximal tree exists by Zorn's lemma (and contains every vertex, otherwise it could be extended). Collapsing T to a point is a homotopy equivalence (a tree is contractible and the pair has the homotopy extension property), and X/T is a wedge of circles, one for each a_α. Apply Theorem 11.4. A finite tree has $E_T = V - 1$ edges, so the number of extra edges is $E - V + 1$. ■

In particular the Euler characteristic $\chi(X) = V - E$ of a finite connected graph is determined by π_1, so it is a homotopy invariant.

14.3 Subgroups of free groups

Theorem 14.4 (Nielsen–Schreier)

Every subgroup H of a free group F is free.

Proof. Realize F as π_1 of a wedge of circles X (a graph). By Theorem 13.4 there is a covering $p : E \to X$ with $p_*\pi_1(E) = H$, and p_* is injective, so $H \cong \pi_1(E)$. By Theorem 14.1, E is a graph, and by Theorem 14.3 its fundamental group is free. ■

Theorem 14.5 (Schreier index formula)

If F is free of finite rank n and $H \subset F$ has finite index k, then H is free of rank $k(n-1)+1$.

Proof. The covering E is a k-sheeted cover of the wedge X of n circles (one vertex, n edges), so E has k vertices and kn edges, and $\operatorname{rank}\pi_1(E) = kn - k + 1$ by Theorem 14.3. ■

Example 14.1

(a) The commutator subgroup of $\mathbb{Z} * \mathbb{Z} = F(a,b)$ is free of infinite rank: it corresponds to the covering of the figure eight by the integer grid in $\mathbb{R}^2$ (the abelianization $\mathbb{Z}^2$ is the deck group), whose fundamental group is free on infinitely many generators.
(b) A free group of rank 2 contains free subgroups of every finite rank and of countably infinite rank; e.g. the kernel of $F(a,b) \to \mathbb{Z}/2$, $a, b \mapsto 1$, has index 2 and rank 3.
(c) In contrast, subgroups of $\mathbb{Z}^n$ are free abelian of rank at most n; for free groups the rank of a finite-index subgroup is larger than that of the group (when $n \geq 2$).